

GSVA for mutil Group

Ximing Ran

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Contents

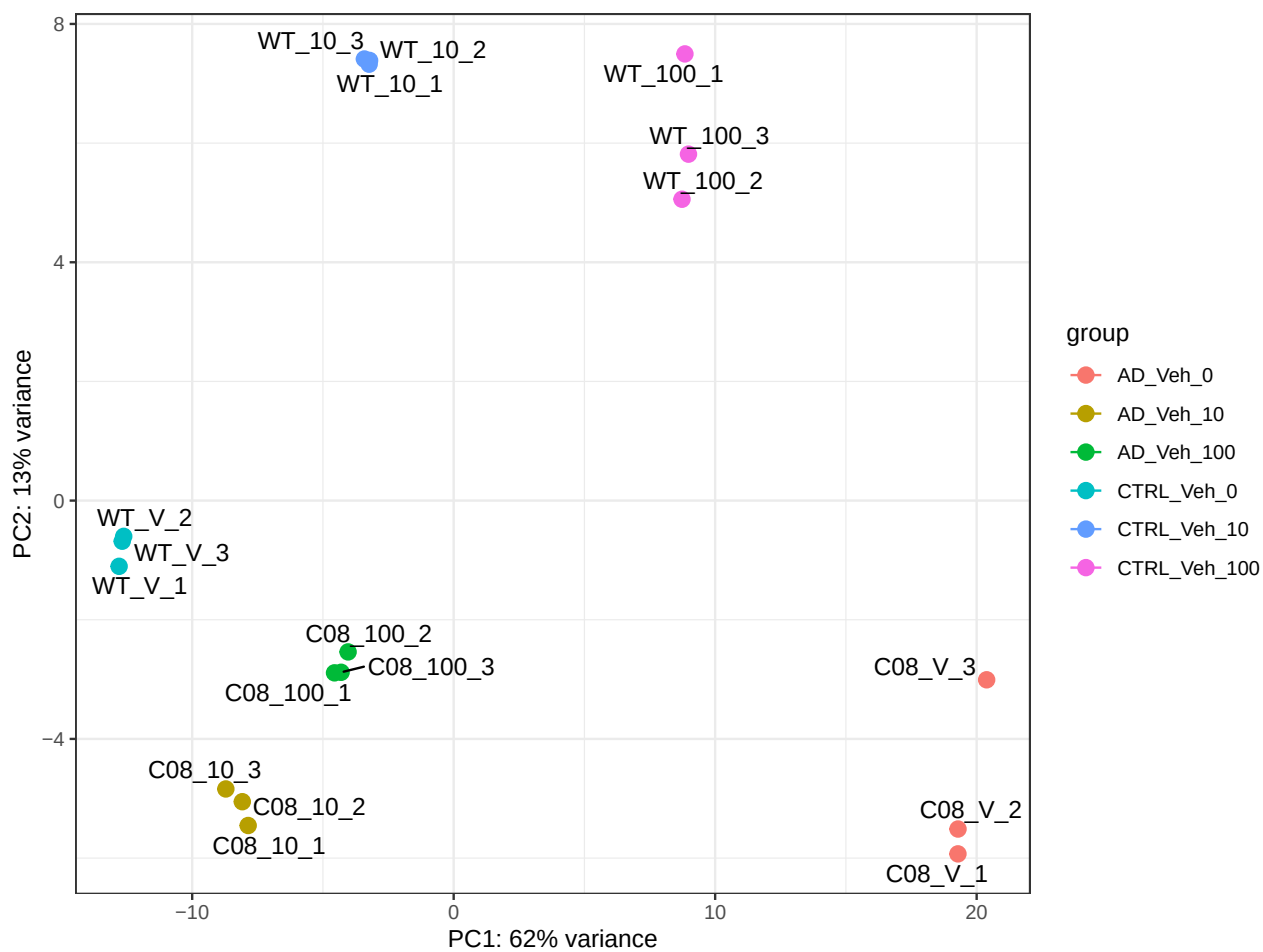
1. Read the count data	2
2. Visualization for reuslt	3
(1) Sample information - PCA plot	3
(2) Sample information - Distance heatmap	4
3. GSEA analysis and BioDomain analysis	5
(1) GSEA analysis for AD vs CTRL	5
(2) GSEA analysis for AD_Drug vs CTRL	5
(3) GSEA analysis for CTRL_Drug vs CTRL	5
4. BioDomain analysis	6
5. Load the GSVA analysis	6
(1) Ridge plot for GSVA score	7
Session information	10

1. Read the count data

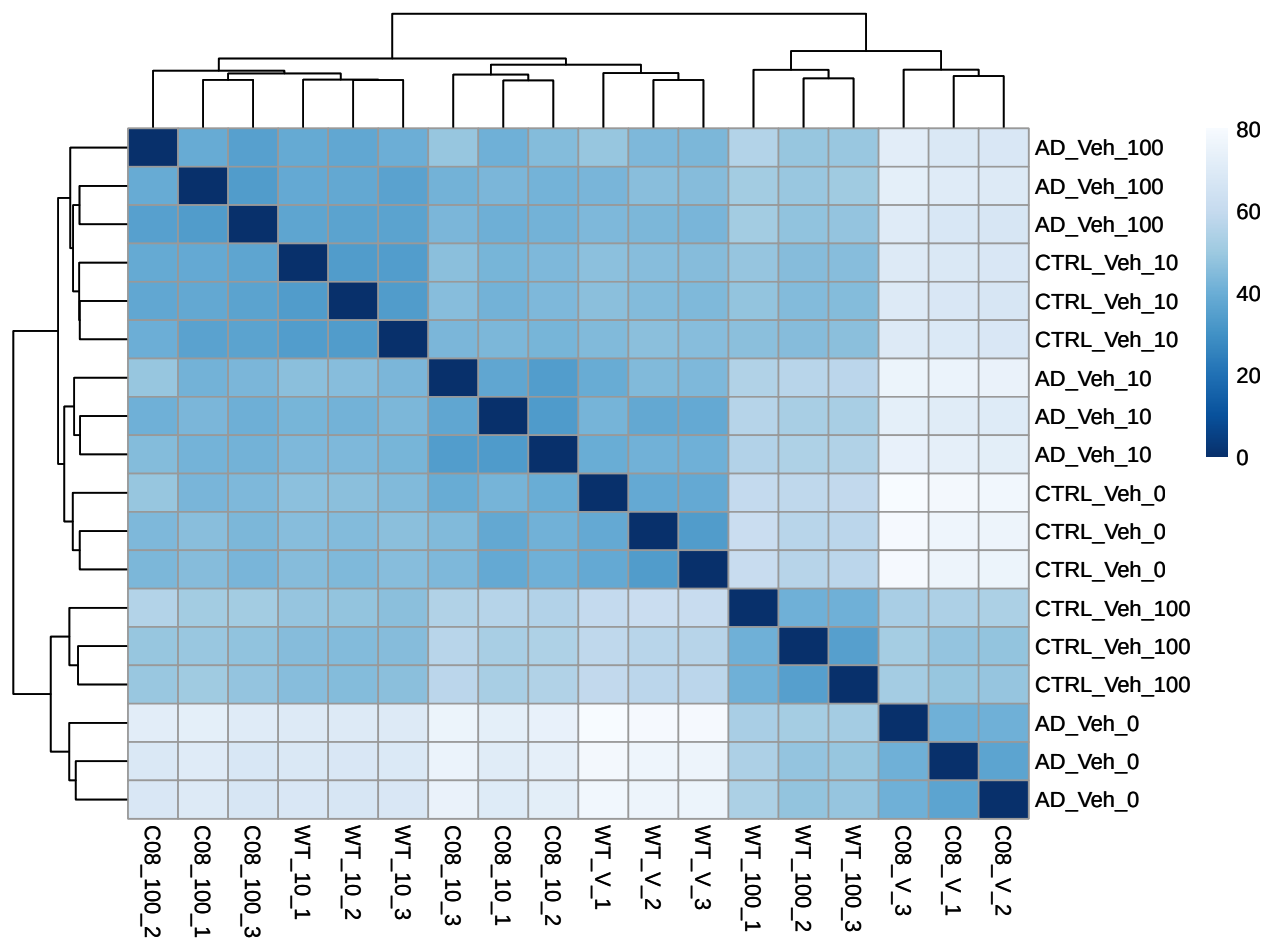
In this section, we will read the clean count data from the `synaptosomes_bulkRNA` folder. We will read the data and merge them into a single table.

2. Visualization for reuslt

(1) Sample information - PCA plot



(2) Sample information - Distance heatmap



3. GSEA analysis and BioDomain analysis

- (1) GSEA analysis for AD vs CTRL
- (2) GSEA analysis for AD_Drug vs CTRL
- (3) GSEA analysis for CTRL_Drug vs CTRL

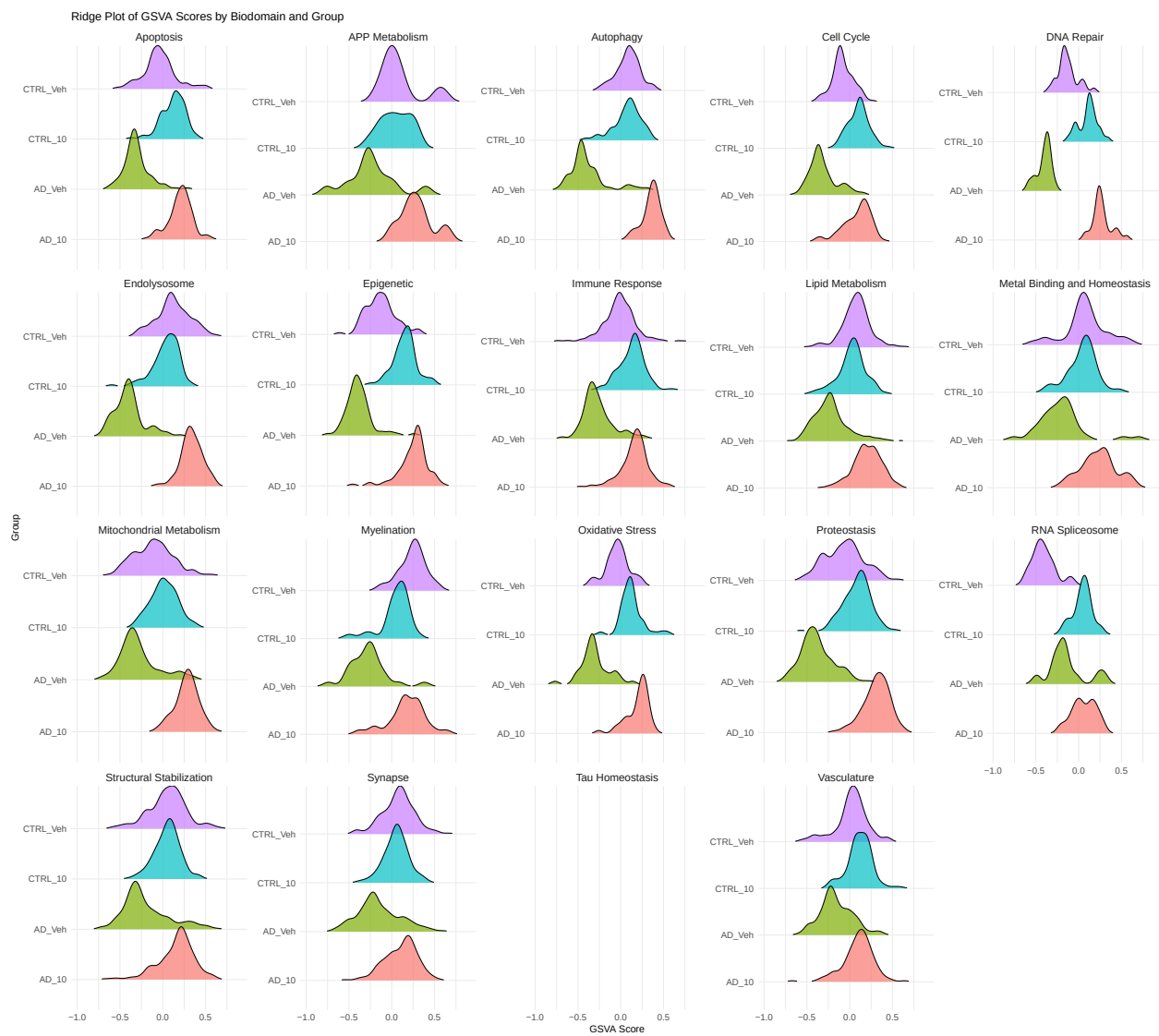
4. BioDomain analysis

5. Load the GSVA analysis

```
## [1] 7406 11
```

```
##          Biodomain      GO_ID      GOterm_Name n_ensGene
## 1 Immune Response GO:0006955      immune response      785
## 2 Immune Response GO:0019882 antigen processing and presentation      151
## 3 Immune Response GO:0002376      immune system process      1215
## 4 Immune Response GO:0045087      innate immune response      799
## 5 Immune Response GO:0006954      inflammatory response      490
## 6 Immune Response GO:0002250      adaptive immune response      733
##  n_entrezGene n_hgncSymbol      ID      AD_Veh_0
## 1          371          488      immune response -0.2967086
## 2           46           46 antigen processing and presentation -0.3979761
## 3          596          849      immune system process      NA
## 4          531          604      innate immune response -0.3117331
## 5          423          420      inflammatory response -0.1975762
## 6          213          468      adaptive immune response -0.2536451
##  AD_Veh_10 CTRL_Veh_0 CTRL_Veh_10
## 1 0.1823246 -0.01628795 0.16955445
## 2 0.2548927 -0.04868701 0.22031646
## 3      NA      NA      NA
## 4 0.1763595 -0.02699089 0.16766773
## 5 0.1769878 0.04226755 0.04721193
## 6 0.1323990 -0.04839786 0.18319579
##
##          AD_Veh_0 AD_Veh_10 CTRL_Veh_0
## GOBP_MITOCHONDRIAL_GENOME_MAINTENANCE -0.3866383 0.3783609 -0.269603842
## GOBP_REPRODUCTION -0.1614877 0.0831761 0.009370062
## GOBP_SINGLE_STRAND_BREAK_REPAIR -0.4296080 0.3919307 0.007202333
## GOBP_REGULATION_OF_DNA_RECOMBINATION -0.3653729 0.2002226 -0.153237594
## GOBP_REGULATION_OF_MITOTIC_RECOMBINATION -0.4406635 0.2510851 -0.120631506
## GOBP_MITOTIC_SPINDLE_ELONGATION -0.3756879 -0.1843761 0.157866075
##          CTRL_Veh_10
## GOBP_MITOCHONDRIAL_GENOME_MAINTENANCE 0.16583963
## GOBP_REPRODUCTION 0.07789206
## GOBP_SINGLE_STRAND_BREAK_REPAIR -0.03092122
## GOBP_REGULATION_OF_DNA_RECOMBINATION 0.18258164
## GOBP_REGULATION_OF_MITOTIC_RECOMBINATION 0.12803435
## GOBP_MITOTIC_SPINDLE_ELONGATION -0.07376122
##
##          ID
## GOBP_MITOCHONDRIAL_GENOME_MAINTENANCE      mitochondrial genome maintenance
## GOBP_REPRODUCTION      reproduction
## GOBP_SINGLE_STRAND_BREAK_REPAIR      single strand break repair
## GOBP_REGULATION_OF_DNA_RECOMBINATION      regulation of dna recombination
## GOBP_REGULATION_OF_MITOTIC_RECOMBINATION      regulation of mitotic recombination
## GOBP_MITOTIC_SPINDLE_ELONGATION      mitotic spindle elongation
```

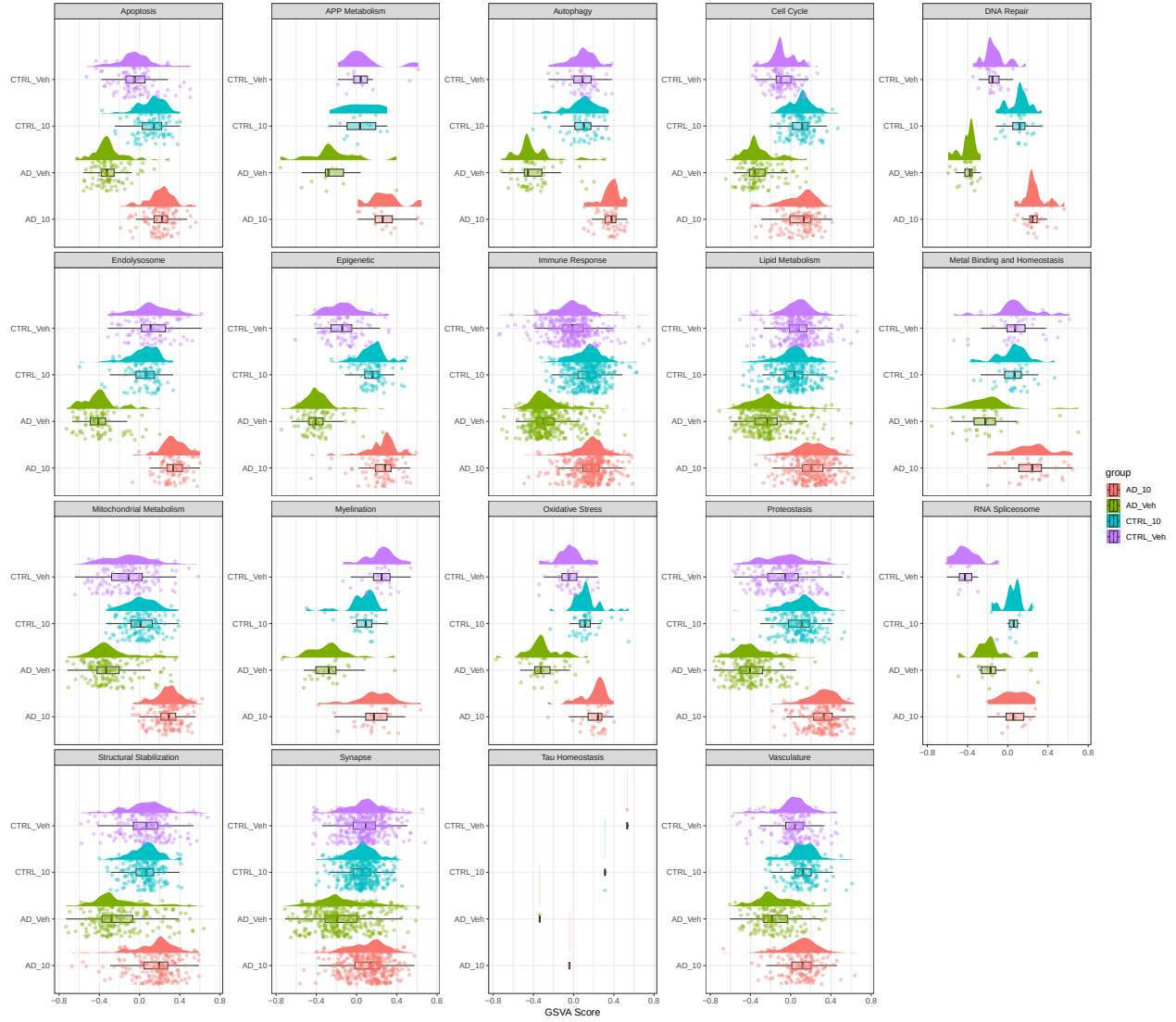
(1) Ridge plot for GSVA score



(2) Raincloud plot for GSVA score

```
## List of 4
## $ axis.text.x      :List of 11
##   ..$ family       : NULL
##   ..$ face         : NULL
##   ..$ colour       : NULL
##   ..$ size         : num 10
##   ..$ hjust        : NULL
##   ..$ vjust        : NULL
##   ..$ angle        : NULL
##   ..$ lineheight   : NULL
##   ..$ margin       : NULL
##   ..$ debug        : NULL
##   ..$ inherit.blank: logi FALSE
##   ..- attr(*, "class")= chr [1:2] "element_text" "element"
## $ axis.text.y      :List of 11
##   ..$ family       : NULL
##   ..$ face         : NULL
##   ..$ colour       : NULL
##   ..$ size         : num 11
##   ..$ hjust        : NULL
##   ..$ vjust        : NULL
##   ..$ angle        : NULL
##   ..$ lineheight   : NULL
##   ..$ margin       : NULL
##   ..$ debug        : NULL
##   ..$ inherit.blank: logi FALSE
##   ..- attr(*, "class")= chr [1:2] "element_text" "element"
## $ legend.position: chr "none"
## $ strip.text      :List of 11
##   ..$ family       : NULL
##   ..$ face         : NULL
##   ..$ colour       : NULL
##   ..$ size         : num 13
##   ..$ hjust        : NULL
##   ..$ vjust        : NULL
##   ..$ angle        : NULL
##   ..$ lineheight   : NULL
##   ..$ margin       : NULL
##   ..$ debug        : NULL
##   ..$ inherit.blank: logi FALSE
##   ..- attr(*, "class")= chr [1:2] "element_text" "element"
## - attr(*, "class")= chr [1:2] "theme" "gg"
## - attr(*, "complete")= logi FALSE
## - attr(*, "validate")= logi TRUE
```

GSVA Distribution per Biodomain (Raincloud Plot)



Session information

```
## R version 4.4.0 (2024-04-24)
## Platform: aarch64-apple-darwin20
## Running under: macOS Sonoma 14.3.1
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib; LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] parallel stats4 stats graphics grDevices utils datasets
## [8] methods base
##
## other attached packages:
## [1] ggdist_3.3.2 ggribes_0.5.6
## [3] GSEABase_1.66.0 graph_1.82.0
## [5] annotate_1.82.0 XML_3.99-0.18
## [7] extrafont_0.19 ggsignif_0.6.4
## [9] patchwork_1.3.0 decoupleR_2.10.0
## [11] GSVA_1.52.3 BiocParallel_1.38.0
## [13] edgeR_4.2.2 limma_3.60.6
## [15] GenomicFeatures_1.56.0 biomaRt_2.60.1
## [17] gprofiler2_0.2.3 RColorBrewer_1.1-3
## [19] data.table_1.16.4 org.Hs.eg.db_3.19.1
## [21] AnnotationDbi_1.66.0 clusterProfiler_4.12.6
## [23] ggfortify_0.4.17 pheatmap_1.0.12
## [25] EnhancedVolcano_1.22.0 ggrepel_0.9.6
## [27] apeglm_1.26.1 DESeq2_1.44.0
## [29] SummarizedExperiment_1.34.0 Biobase_2.64.0
## [31] MatrixGenerics_1.16.0 matrixStats_1.5.0
## [33] reshape2_1.4.4 Matrix_1.7-2
## [35] Signac_1.14.0 Seurat_5.2.1
## [37] SeuratObject_5.0.2 sp_2.2-0
## [39] rtracklayer_1.64.0 GenomicRanges_1.56.2
## [41] GenomeInfoDb_1.40.1 IRanges_2.38.1
## [43] S4Vectors_0.42.1 BiocGenerics_0.50.0
## [45] knitr_1.49 lubridate_1.9.4
## [47] forcats_1.0.0 stringr_1.5.1
## [49] dplyr_1.1.4 purrr_1.0.4
## [51] readr_2.1.5 tidyr_1.3.1
## [53] tibble_3.2.1 ggplot2_3.5.1
## [55] tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] SpatialExperiment_1.14.0 R.methodsS3_1.8.2
## [3] progress_1.2.3 goftest_1.2-3
## [5] HDF5Array_1.32.1 Biostrings_2.72.1
```

## [7] vctr_0.6.5	spatstat.random_3.3-2
## [9] digest_0.6.37	png_0.1-8
## [11] deldir_2.0-4	parallelly_1.42.0
## [13] magick_2.8.5	MASS_7.3-64
## [15] httpuv_1.6.15	qvalue_2.36.0
## [17] withr_3.0.2	xfun_0.51
## [19] ggfun_0.1.8	survival_3.8-3
## [21] memoise_2.0.1	gson_0.1.0
## [23] systemfonts_1.2.1	ragg_1.3.3
## [25] tidytree_0.4.6	zoo_1.8-12
## [27] pbapply_1.7-2	R.oo_1.27.0
## [29] prettyunits_1.2.0	KEGGREST_1.44.1
## [31] promises_1.3.2	httr_1.4.7
## [33] restfulr_0.0.15	rhdf5filters_1.16.0
## [35] globals_0.16.3	fitdistrplus_1.2-2
## [37] rhdf5_2.48.0	rstudioapi_0.17.1
## [39] UCSC.utils_1.0.0	miniUI_0.1.1.1
## [41] generics_0.1.3	DOSE_3.30.5
## [43] curl_6.2.1	zlibbioc_1.50.0
## [45] ScaledMatrix_1.12.0	ggraph_2.2.1
## [47] polyclip_1.10-7	GenomeInfoDbData_1.2.12
## [49] SparseArray_1.4.8	xtable_1.8-4
## [51] evaluate_1.0.3	S4Arrays_1.4.1
## [53] BiocFileCache_2.12.0	hms_1.1.3
## [55] irlba_2.3.5.1	colorspace_2.1-1
## [57] filelock_1.0.3	ROCR_1.0-11
## [59] reticulate_1.40.0	spatstat.data_3.1-4
## [61] magrittr_2.0.3	lmtest_0.9-40
## [63] later_1.4.1	viridis_0.6.5
## [65] ggtree_3.12.0	lattice_0.22-6
## [67] spatstat.geom_3.3-5	future.apply_1.11.3
## [69] scattermore_1.2	shadowtext_0.1.4
## [71] cowplot_1.1.3	RcppAnnoy_0.0.22
## [73] pillar_1.10.1	nlme_3.1-167
## [75] compiler_4.4.0	beachmat_2.20.0
## [77] RSpectra_0.16-2	stringi_1.8.4
## [79] tensor_1.5	GenomicAlignments_1.40.0
## [81] plyr_1.8.9	crayon_1.5.3
## [83] abind_1.4-8	BiocIO_1.14.0
## [85] gridGraphics_0.5-1	emdbbook_1.3.13
## [87] locfit_1.5-9.11	graphlayouts_1.2.2
## [89] bit_4.5.0.1	fastmatch_1.1-6
## [91] textshaping_1.0.0	codetools_0.2-20
## [93] BiocSingular_1.20.0	plotly_4.10.4
## [95] mime_0.12	splines_4.4.0
## [97] Rcpp_1.0.14	fastDummies_1.7.5
## [99] sparseMatrixStats_1.16.0	dbplyr_2.5.0
## [101] Rttf2pt1_1.3.12	blob_1.2.4
## [103] here_1.0.1	fs_1.6.5
## [105] listenv_0.9.1	ggplotify_0.1.2
## [107] statmod_1.5.0	tzdb_0.4.0
## [109] tweenr_2.0.3	pkgconfig_2.0.3
## [111] tools_4.4.0	cachem_1.1.0
## [113] RSQLite_2.3.9	viridisLite_0.4.2

## [115] DBI_1.2.3	numDeriv_2016.8-1.1
## [117] fastmap_1.2.0	rmarkdown_2.29
## [119] scales_1.3.0	grid_4.4.0
## [121] ica_1.0-3	Rsamtools_2.20.0
## [123] coda_0.19-4.1	dotCall64_1.2
## [125] RANN_2.6.2	farver_2.1.2
## [127] tidygraph_1.3.1	scatterpie_0.2.4
## [129] yaml_2.3.10	cli_3.6.4
## [131] lifecycle_1.0.4	uwot_0.2.2
## [133] mvtnorm_1.3-3	timechange_0.3.0
## [135] gtable_0.3.6	rjson_0.2.23
## [137] progressr_0.15.1	ape_5.8-1
## [139] jsonlite_1.9.0	RcppHNSW_0.6.0
## [141] bitops_1.0-9	bit64_4.6.0-1
## [143] Rtsne_0.17	yulab.utils_0.2.0
## [145] spatstat.utils_3.1-2	bdsmatrix_1.3-7
## [147] GOSemSim_2.30.2	distributional_0.5.0
## [149] spatstat.univar_3.1-1	R.utils_2.12.3
## [151] lazyeval_0.2.2	shiny_1.10.0
## [153] htmltools_0.5.8.1	enrichplot_1.24.4
## [155] GO.db_3.19.1	sctransform_0.4.1
## [157] rappdirs_0.3.3	tinytex_0.55
## [159] glue_1.8.0	spam_2.11-1
## [161] httr2_1.1.0	XVector_0.44.0
## [163] RCurl_1.98-1.16	rprojroot_2.0.4
## [165] treeio_1.28.0	gridExtra_2.3
## [167] extrafontdb_1.0	igraph_2.1.4
## [169] R6_2.6.1	SingleCellExperiment_1.26.0
## [171] labeling_0.4.3	RcppRoll_0.3.1
## [173] cluster_2.1.8	bbmle_1.0.25.1
## [175] Rhdf5lib_1.26.0	aplot_0.2.4
## [177] DelayedArray_0.30.1	tidyselect_1.2.1
## [179] ggforce_0.4.2	xml2_1.3.6
## [181] future_1.34.0	rsvd_1.0.5
## [183] munsell_0.5.1	KernSmooth_2.23-26
## [185] htmlwidgets_1.6.4	fgsea_1.30.0
## [187] rlang_1.1.5	spatstat.sparse_3.1-0
## [189] spatstat.explore_3.3-4	